

CREDIT RISK ASSESSMENT: PREDICTING CREDIT CARD DEFAULT USING CUSTOMER AND PAYMENT CHARACTERISTICS

Project Report Submitted in fulfillment of the requirements for the Data Science & AI Training Program by
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Data Science & AI

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DECLARATION

This section is prepared as a submission declaration for the capstone report. Before submission, the student should review the statement and confirm that it accurately reflects their participation, review of the analysis, and the academic-integrity requirements of the programme.

Student confirmation: I have reviewed the analysis, results, references and supporting notebook included with this report and accept responsibility for the final submitted work.

ACKNOWLEDGEMENT

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ABSTRACT

This project develops a data-driven classification approach for predicting whether a credit card client will default on the following month's payment. The analysis uses the supplied P_5_UCI_Credit_Card.csv dataset, which contains 30,000 customer records and 23 explanatory variables covering credit limits, demographic characteristics, repayment status, bill amounts and previous payments. Data-quality checks found no missing values and no duplicate rows. Category codes were harmonised and additional aggregate features were engineered to summarise historical bills, payments, utilisation and repayment delay. Exploratory and statistical analysis showed that recent repayment status was strongly associated with the observed default outcome, while defaulting accounts also had lower average credit limits and lower aggregate payment amounts in this sample. Three classification models were evaluated using a stratified 80/20 train-test split: Logistic Regression, Decision Tree and Random Forest. The Random Forest achieved the highest test accuracy (0.779), F1-score (0.538) and ROC-AUC (0.778) among the tested models. The results indicate that repayment-history variables provide substantial predictive information, while the model should be treated as a decision-support tool rather than evidence of causation or a complete credit-risk assessment.

Keywords: Credit Risk; Credit Card Default; Classification; Exploratory Data Analysis; Random Forest; Predictive Analytics

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1. INTRODUCTION

1.1 Background

Credit risk assessment is a classification problem in which information available about a customer is used to estimate the likelihood of a future adverse repayment outcome. Predictive models can support risk monitoring and resource allocation when combined with institutional policy and human oversight.

1.2 Context of the Problem

The supplied dataset records credit limits, demographic characteristics, six months of repayment status, six months of bill statements and six months of previous payments. The target is default on the following month's payment. The UCI repository describes this as the Default of Credit Card Clients dataset with 30,000 instances and 23 explanatory features.

1.3 Scope of the Study

This study evaluates binary classification models using the supplied variables. It does not claim causal effects or production-level validation on an external population.

2. PROBLEM STATEMENT AND MOTIVATION

Can customer characteristics, credit limits, repayment history, bill amounts and previous payments be used to predict whether a credit-card client will default on the following month's payment?

3. OBJECTIVES

1. Assess data structure and quality. 2. Identify important patterns through EDA and descriptive statistics. 3. Prepare the data with justified preprocessing and feature engineering. 4. Build and compare three classification models. 5. Evaluate performance and translate findings into practical recommendations.

4. RESEARCH QUESTIONS

1. What are the main data-quality characteristics? 2. How does default vary with repayment status and payment behaviour? 3. Which variables contribute most to prediction? 4. How accurately can the tested models predict unseen outcomes? 5. What limitations should be considered?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The supplied file is P_5_UCI_Credit_Card.csv. Its structure matches the UCI Default of Credit Card Clients dataset. UCI reports 30,000 instances, 23 features, a binary classification task and no missing values.

5.2 Dataset Size and Structure

Item	Result
Observations	30,000
Columns in supplied CSV	25
Predictor variables documented by UCI	23
Identifier	ID
Target	default.payment.next.month
Default cases	6,636 (22.12%)
Non-default cases	23,364 (77.88%)
Missing values	0

Duplicate rows	0
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Table 1. Summary of the supplied dataset.

5.3 Data Dictionary

Variable / group	Meaning	Type	Role
ID	Client identifier	Integer	Identifier
LIMIT_BAL	Amount of given credit	Numeric	Feature
SEX	Gender code	Categorical	Feature
EDUCATION	Education level code	Categorical	Feature
MARRIAGE	Marital-status code	Categorical	Feature
AGE	Age in years	Numeric	Feature
PAY_0, PAY_2...PAY_6	Repayment status across six months	Ordinal/numeric code	Feature
BILL_AMT1...BILL_AMT6	Monthly bill statement amounts	Numeric	Feature
PAY_AMT1...PAY_AMT6	Monthly previous payment amounts	Numeric	Feature
default.payment.next.month	Default payment next month (0/1)	Binary	Target

Table 2. Compact data dictionary.

6. DATA CLEANING

6.1 Missing Values

The raw file contains 0 missing cells; no imputation was necessary.

6.2 Duplicate Records

The dataset contains 0 exact duplicate rows.

6.3 Data Types and Inconsistent Values

All columns loaded as numeric. Education codes 0, 5 and 6 were grouped into 'other' for modelling; marriage code 0 was grouped into 'other'.

6.4 Outlier Handling

Large monetary observations were retained because they were not treated as obvious data-entry errors.

6.5 Before vs After Cleaning

Quality check	Before	Action	After
Missing cells	0	No imputation required	0
Duplicate rows	0	No removal required	0
Education special codes	0, 5, 6 present	Grouped as other for modelling	Harmonised
Marriage code 0	Present	Grouped as other for modelling	Harmonised

Table 3. Data-quality checks and treatment decisions.

7. DATA PREPROCESSING

Categorical variables were one-hot encoded. Numerical variables were standardised inside a ColumnTransformer pipeline. The data was split 80/20 using stratification and random_state=42. Engineered variables include average and total bills/payments, latest utilisation, maximum delay and average delay.

8. EXPLORATORY DATA ANALYSIS

The target distribution is moderately imbalanced, with 22.12% default observations. Recent repayment status shows a strong association with the default outcome. Credit-limit distributions differ across outcome groups, and monthly bill variables are correlated across adjacent months.

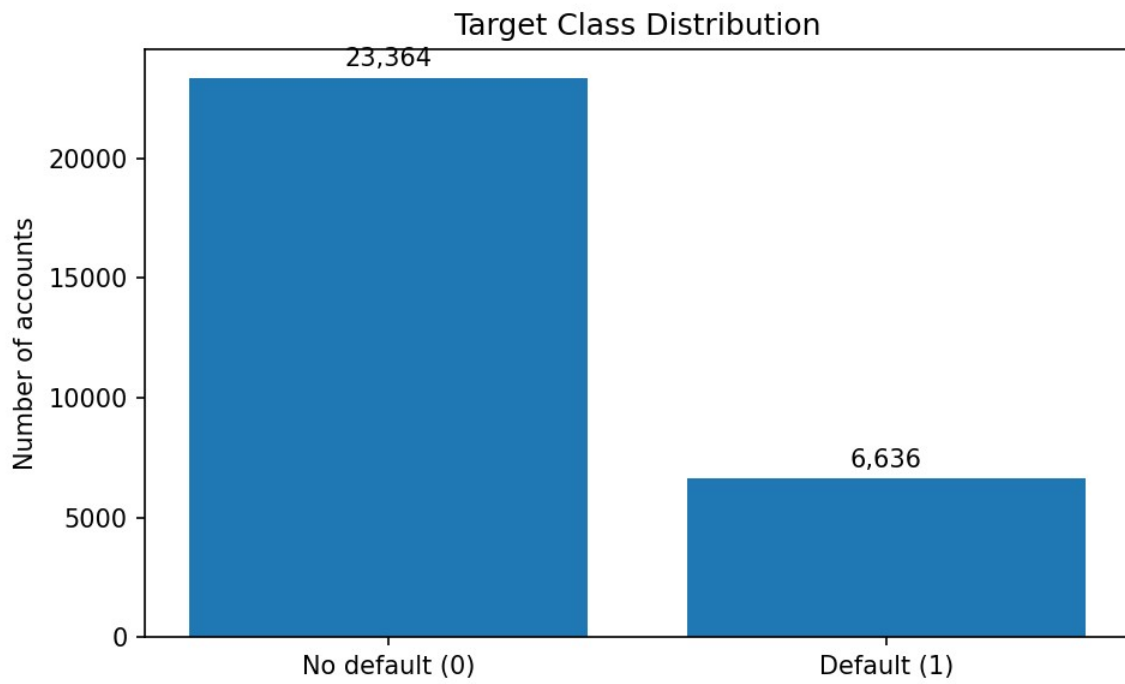


Figure 1. Target class distribution.

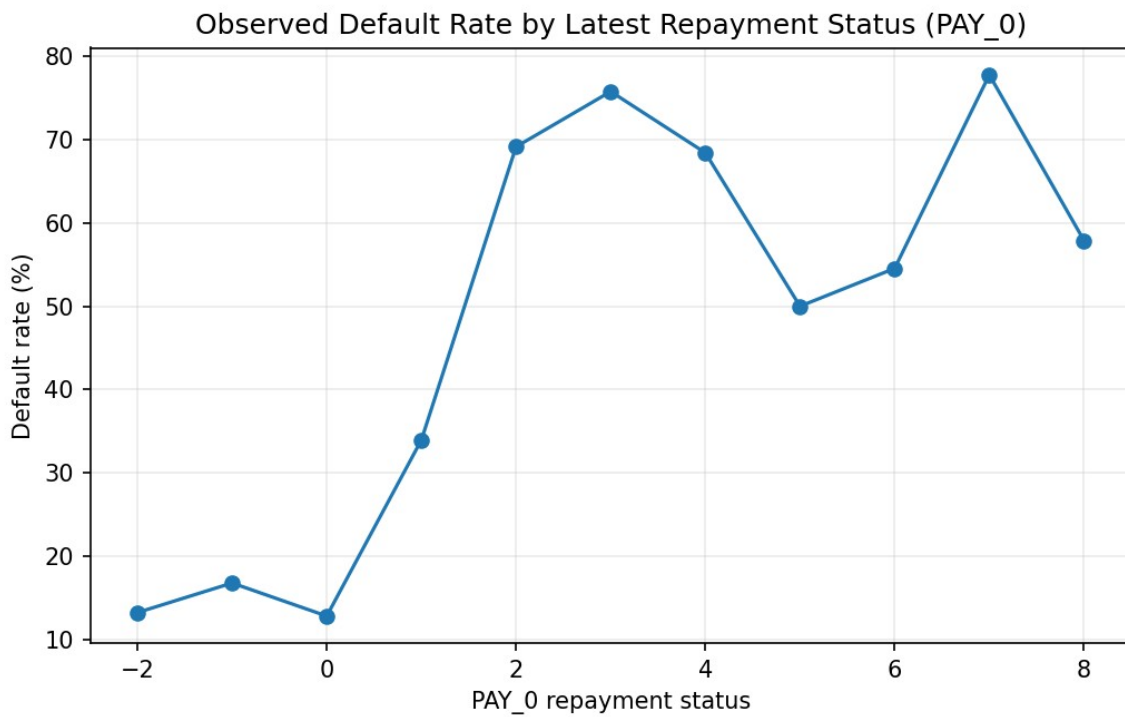


Figure 2. Default rate by latest repayment status.

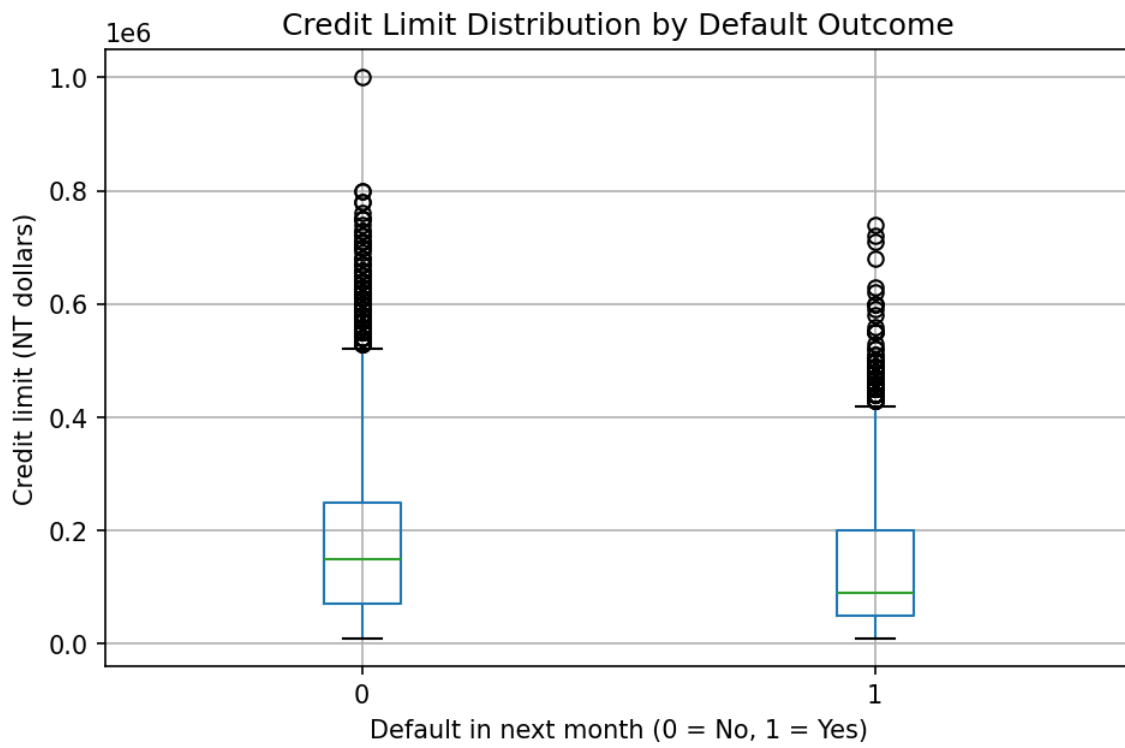


Figure 3. Credit-limit distribution by outcome.

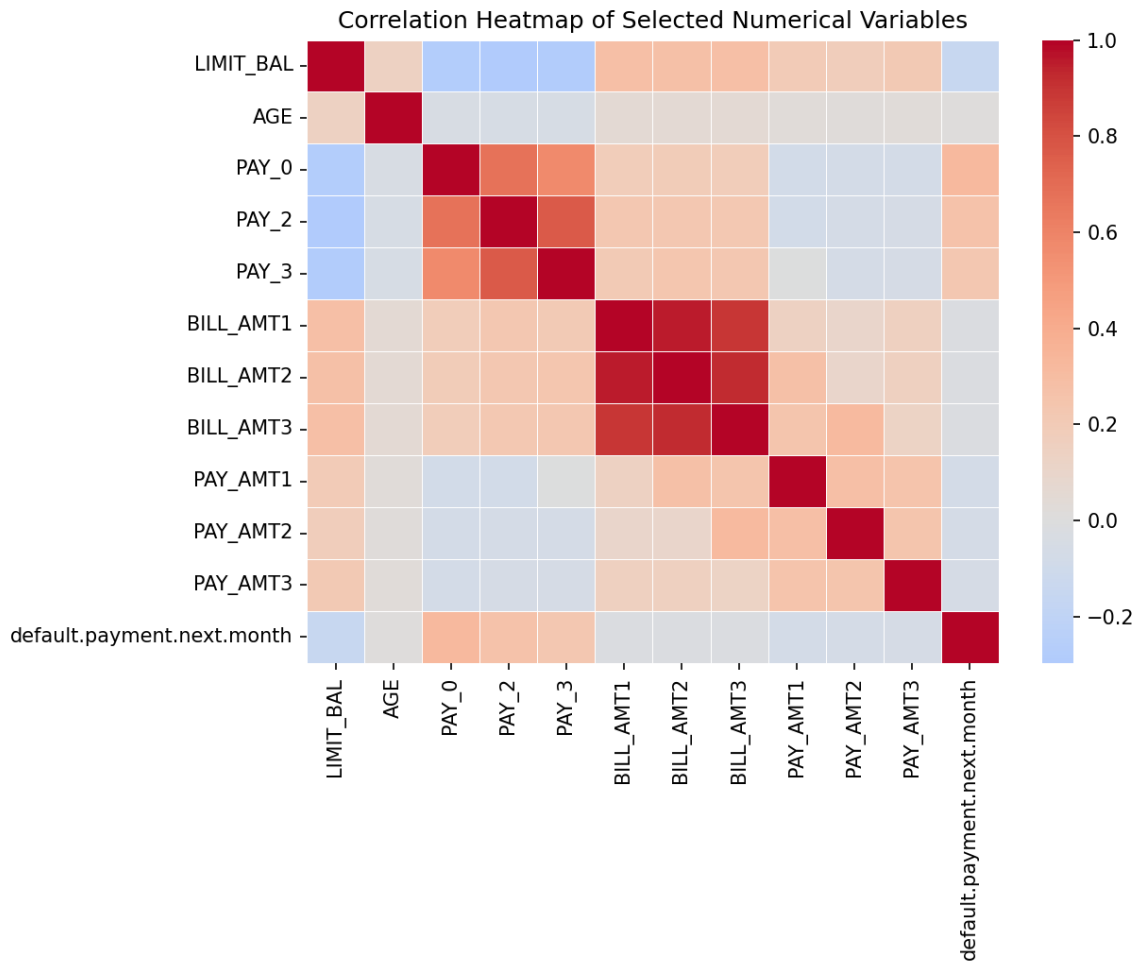


Figure 4. Correlation heatmap.

9. STATISTICAL ANALYSIS

Measure	Non-default mean	Default mean	Interpretation
Credit limit (mean)	178,099.73	130,109.66	
Age (mean)	35.42	35.73	
Average bill amount (mean)	45,404.82	43,470.49	
Average payment amount (mean)	5,828.24	3,328.22	
Latest-period utilisation (mean)	0.40	0.49	
Maximum repayment-delay code (mean)	0.20	1.27	
Total six-month payments (mean)	34,969.42	19,969.29	

Table 4. Descriptive statistics of selected variables.

Defaulting observations have a lower mean credit limit (130,110) than non-defaulting observations (178,100) and higher repayment-delay summaries. These are descriptive associations, not causal effects.

10. FEATURE ENGINEERING

Feature	Construction	Purpose
AVG_BILL_AMT	Mean of BILL_AMT1...BILL_AMT6	Summarise recent bill burden
AVG_PAY_AMT	Mean of PAY_AMT1...PAY_AMT6	Summarise recent payment behaviour
TOTAL_BILL_AMT	Sum of six bill amounts	Capture cumulative billed amount
TOTAL_PAY_AMT	Sum of six payment amounts	Capture cumulative payments
UTILIZATION_1	BILL_AMT1 / LIMIT_BAL	Approximate latest-period credit utilisation
MAX_PAY_DELAY	Maximum of six repayment-status codes	Capture worst recent delay
AVG_PAY_DELAY	Mean of six repayment-status codes	Capture average recent delay

Table 5. Engineered features.

11. MODEL DEVELOPMENT

The task is binary classification with default.payment.next.month as the target. Logistic Regression, Decision Tree and Random Forest were evaluated. Class-weight balancing was used and preprocessing was placed inside reproducible pipelines.

12. MODEL EVALUATION

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.723	0.417	0.638	0.504	0.727
Decision Tree	0.763	0.473	0.616	0.535	0.769
Random Forest	0.779	0.501	0.582	0.538	0.778

Table 6. Test-set model performance.

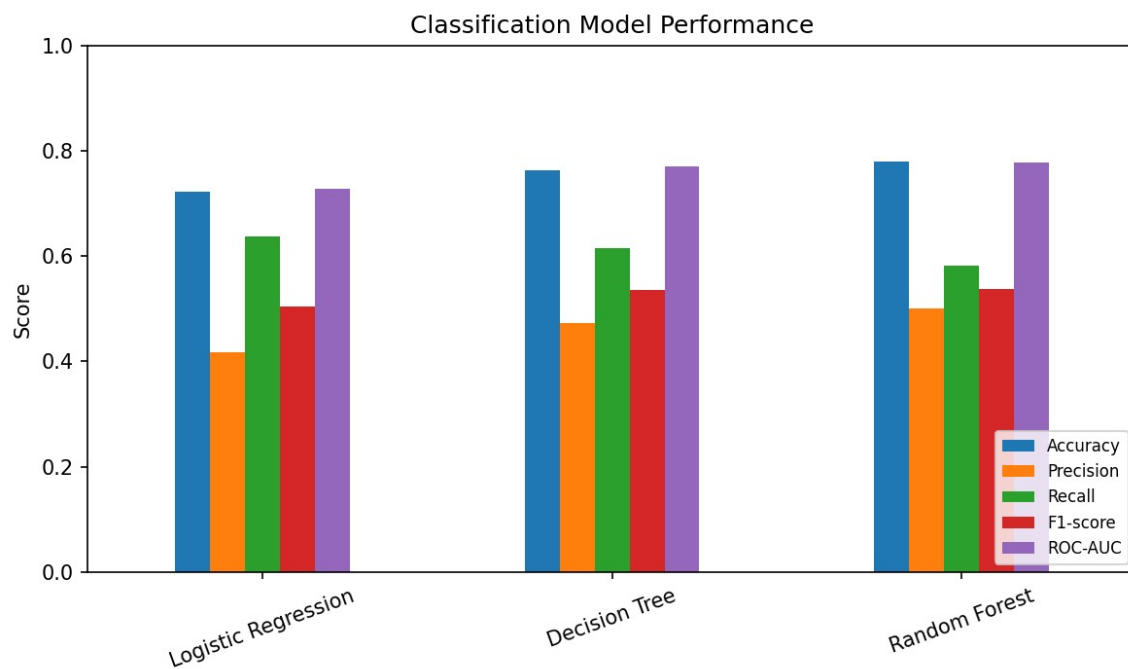


Figure 6. Classification model comparison.

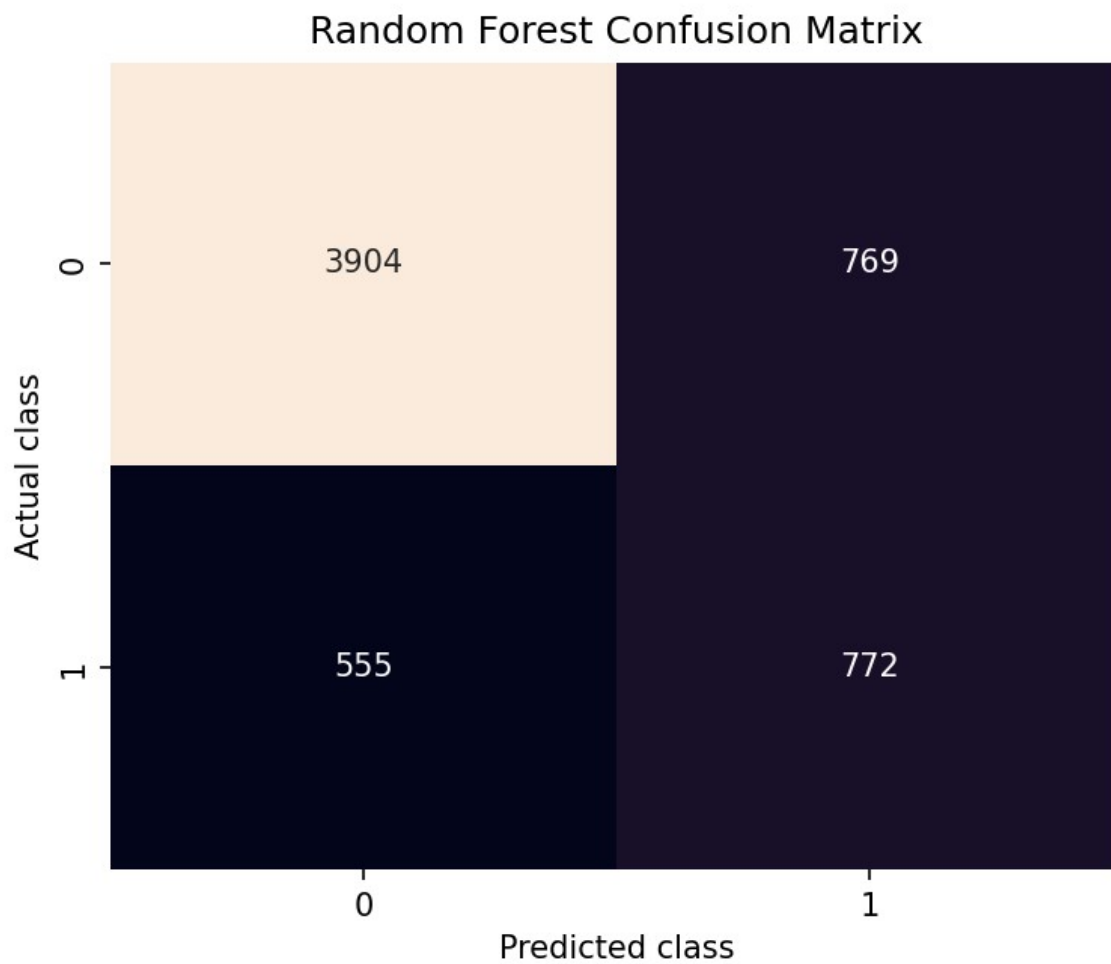


Figure 5. Random Forest confusion matrix.

Random Forest achieved accuracy 0.779, precision 0.501, recall 0.582, F1-score 0.538, and ROC-AUC 0.778 on the held-out test set.

13. RESULTS AND FINDINGS

1. The supplied dataset is complete: it contains 30,000 observations with no missing cells and no exact duplicate rows.
2. The default class represents 22.12% of the sample, so accuracy alone is insufficient for judging the classifiers.
3. Recent repayment status is strongly associated with default. For example, the observed default rate rises from about 12.8% at PAY_0=0 to about 69.1% at PAY_0=2.
4. The Random Forest achieved the highest test F1-score (0.538) and ROC-AUC (0.778) among the tested models.
5. Random Forest feature importance places repayment-delay variables near the top; MAX_PAY_DELAY, PAY_0 and AVG_PAY_DELAY are the three largest importance values in the fitted model.

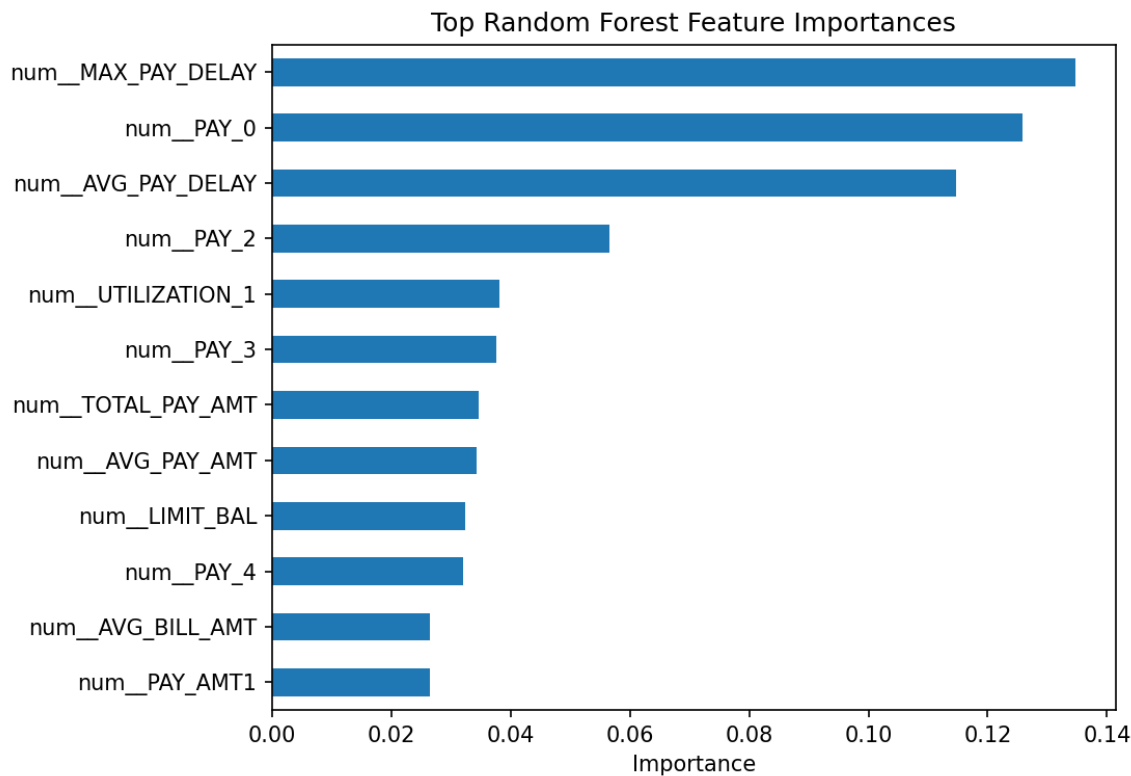


Figure 7. Top Random Forest feature importances.

14. DISCUSSION

Recent repayment behaviour is the strongest source of predictive information in this experiment. The Random Forest provides the strongest overall combination of accuracy, F1-score and ROC-AUC among the tested models, but this result is specific to the chosen dataset, split and feature pipeline.

15. LIMITATIONS

- Historical population and period may limit generalisation.
- Important risk variables not present in the dataset cannot be modelled.
- The cost of false positives versus false negatives is not specified.
- A single train-test split is weaker evidence than cross-validation plus external validation.
- Feature importance does not establish causation.

16. CONCLUSION

The project demonstrates a complete credit-card default classification workflow. The Random Forest produced the strongest held-out performance among the tested models, while repayment-delay variables were the most prominent predictive signals. The model should be treated as decision support and validated carefully before any operational use.

17. RECOMMENDATIONS AND FUTURE WORK

- Use repayment-history variables as an important risk-monitoring input.
- Report recall, precision and F1-score alongside accuracy.
- Combine model output with policy, human review and additional verified information.
- Monitor false negatives explicitly.
- Use cross-validation, calibration, threshold optimisation and external validation in future work.

18. REFERENCES

1. Yeh, I.-C. (2009). Default of Credit Card Clients. UCI Machine Learning Repository. DOI: 10.24432/C55S3H. [NciteÖturn0search0Ô](#)
2. Yeh, I.-C., & Lien, C.-H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. Expert Systems with Applications. [NciteÖturn0search0Ô](#)
3. India Data Research Academy. (2026). IDRA Capstone Project Guide 2026 and IDRA Capstone Project Template Sample. Supplied training materials.

APPENDIX A. CODE

The complete reproducible code is in `Credit_Risk_Capstone_Analysis.ipynb`, covering loading, cleaning, preprocessing, EDA, statistical analysis, modelling and evaluation.

APPENDIX B. ADDITIONAL FIGURES AND TABLES

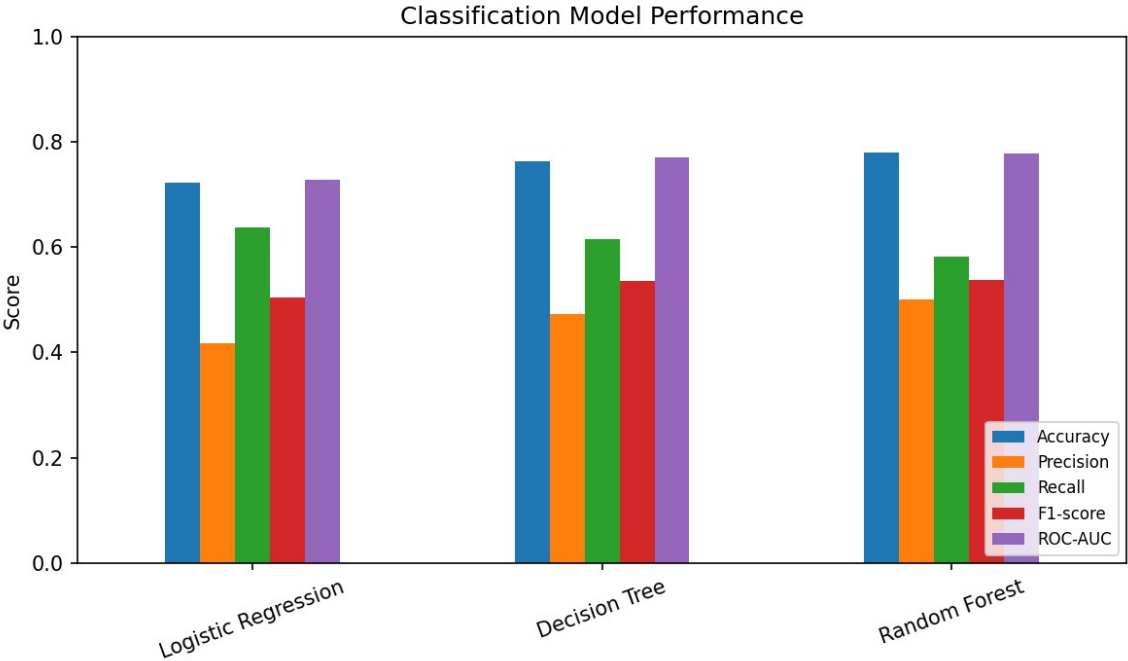


Figure B1. Additional model comparison chart.